

TRUSTOS

Enterprise AI-Native Trust & Safety Operating System

From Detection to Decision

Autonomous Trust & Safety with Human Oversight

EXECUTIVE WHITEPAPER

CONFIDENTIAL

Author

Aniruddha Vanshiv

Trust & Safety | Risk | Customer Experience

Executive Summary

Trust and Safety has become one of the most consequential operational domains for any enterprise running a digital marketplace, social platform, or collaborative ecosystem. As platform scale grows, so does the complexity of detecting, investigating, and adjudicating violations — from counterfeit listings and fraud to brand abuse, policy violations, and harmful content. Manual review queues overflow. Reactive systems generate false positives at scale. Regulatory pressure mounts. And executive visibility into the state of platform integrity remains dangerously limited.

TrustOS addresses this challenge head-on by providing a unified, AI-orchestrated investigation platform that connects detection signals, policy knowledge, brand authentication, and autonomous risk analysis into a single, governed decision workflow. At its core, TrustOS leverages a multi-agent AI framework — orchestrated through n8n — to perform the investigative work that previously required teams of human analysts, while preserving meaningful human oversight at every critical decision point.

Detect

Surface policy violations and risk signals across marketplace events in real time, before harm escalates.

Investigate

Autonomously gather evidence, apply policy, and authenticate brand claims through specialized AI agents.

Decide

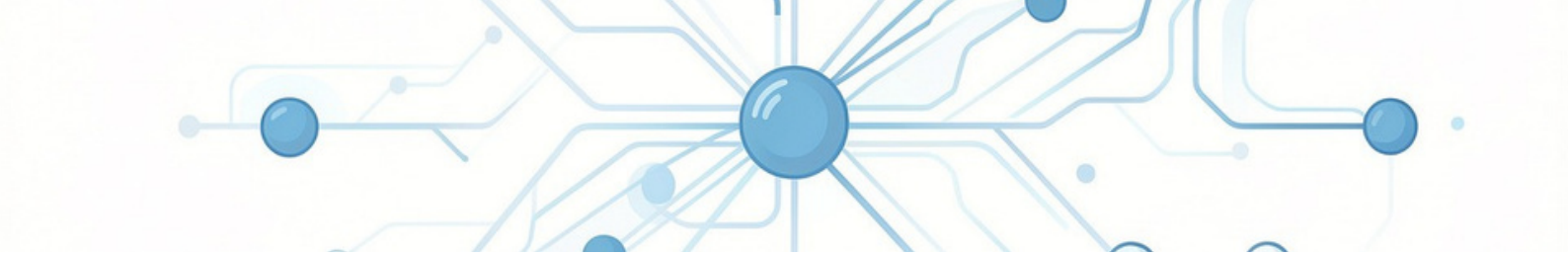
Deliver explainable, auditable compliance decisions via a governed Autonomous Risk Council with human escalation paths.

Report

Provide executive-grade visibility into platform integrity, trend analytics, and regulatory audit logs.

TrustOS is designed for organizations that cannot afford to choose between speed and accuracy, between automation and accountability. It delivers both — at scale, with governance, and with the explainability that enterprise and regulatory stakeholders demand.

-  TrustOS is built on the principle that AI should amplify human judgment, not replace it. Every autonomous decision remains auditable, contestable, and supervised.



TrustOS at a Glance

WHAT IS TRUSTOS

TrustOS is an enterprise AI-powered investigation platform that transforms fragmented Trust & Safety workflows into a unified, explainable, and auditable operating system. It combines specialized AI agents, human oversight, and enterprise governance to deliver compliance-grade decisions at scale.

WHO IS IT FOR



Enterprise Marketplaces



E-commerce Platforms



Digital Platforms



Trust & Safety Teams



Risk & Compliance Teams

CORE CAPABILITIES



Multi-Agent AI Investigation



Explainable Decision Making



Human-in-the-Loop Governance



Executive Analytics



Enterprise Auditability

TECHNOLOGY STACK



n8n (Workflow Orchestration)



OpenAI Models (AI Intelligence)



Google Sheets (Operational Data Store)



Multi-Agent Investigation Framework



Executive Analytics Dashboard

CURRENT IMPLEMENTATION

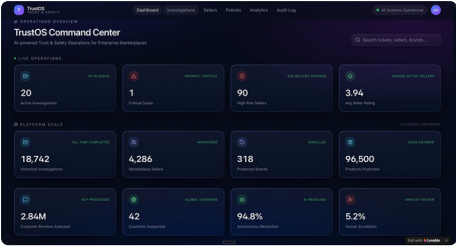
- Dashboard
- Investigation Queue
- Investigation Detail
- Sellers Intelligence
- Policies Module
- AI Investigation Progress
- Analytics Dashboard
- Audit Log
- Multi-Agent Workflow
- n8n Orchestration
- OpenAI-powered AI Agents
- Google Sheets Backend

FUTURE VISION

TrustOS is designed as an evolving platform. The roadmap includes expanded AI capabilities, regulatory intelligence, advanced threat detection, and deeper enterprise integrations—positioning Trust & Safety as a strategic, intelligence-driven function rather than a reactive moderation service.

Platform Overview

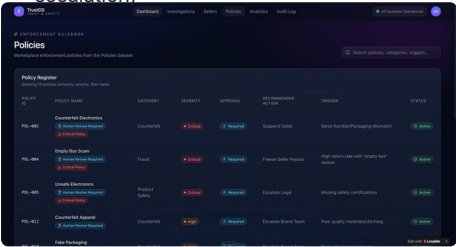
TrustOS is aunified investigationplatformintegratingseller intelligence, policy management, multi-agent AI, and executive analytics. Below is a visual tour of the complete product.



Dashbo ard

Executive command center for real-time platform monitoring

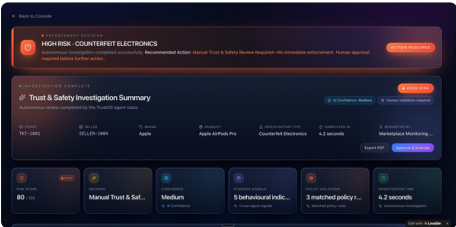
- Live operational metrics (20 active investigations, 1 critical case)
- Real-time KPIs (18,742 listings processed, 4,286 sellers reviewed)
- Investigation health indicators (94.8% accuracy, 5.2% escalation)



Policies

Centralized policy repository and enforcement rules

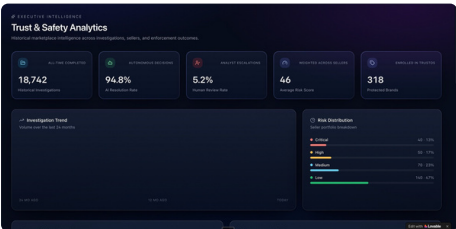
- Policy categories (Counterfeit Electronics, Fake Packaging, Unsafe Electronics)
- Enforcement rules with approval workflows
- Policy status tracking (Active, Inactive, Pending)



Investigation Detail

AI investigation output with explainable reasoning

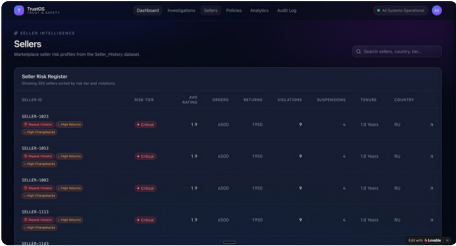
- Automated Trust & Safety Review with AI agent findings
- Risk assessment and confidence scoring
- Manual review escalation options



Anal ytics

Executive reporting and operational insights

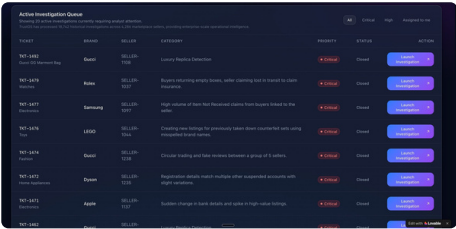
- Executive KPIs (18,742 investigations, 94.8% accuracy, 5.2% escalation)
- Investigation trends and violation patterns
- Risk distribution analysis by category



Sellers Intelligence

Comprehensive seller risk profiling and behavioral analysis

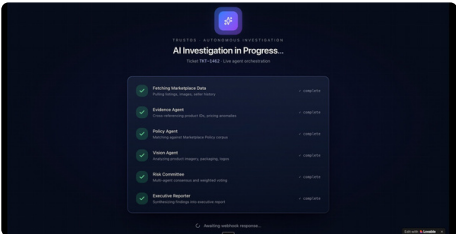
- Seller risk classification (Critical, High, Medium, Low)
- Historical metrics (orders, returns, violations, suspensions)
- Behavioral patterns and marketplace activity tracking



Investigation Queue

Prioritized investigation workflow with SLA tracking

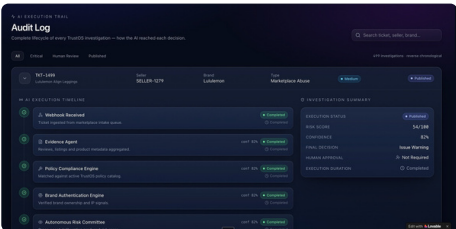
- Prioritized investigation list with risk scoring
- SLA tracking and status indicators
- Category-based organization and one-click actions



AI Investigation Progress

Real-time multi-agent execution monitoring

- Real-time agent execution status with progress indicators
- Evidence Agent, Policy Agent, Vision Agent execution
- Risk Council and Executive Reporter consolidation



Audit Log

Complete compliance evidence and traceability

- Full investigation traceability with complete history
- Agent execution records and timestamp tracking
- Compliance status indicators and regulatory evidence

The Problem Statement

Enterprise Trust & Safety is facing a structural crisis: content volume, transaction volume, and policy complexity have outpaced what human review teams can handle. Without enterprise-grade automation and governance, the operating model breaks down.

THE CORE CHALLENGES



Scale Outpaces Capacity

Volume of content, listings, and transactions exceeds human review capacity.



Fragmented Tools

Point solutions address single dimensions but lack end-to-end workflow.



Inconsistent Decisions

Manual review leads to inconsistent policy enforcement and appeal friction.



Limited Visibility

Executives lack real-time insight into investigation patterns and risk trends.



Audit & Compliance Risk

Decisions lack explainability and traceability for regulatory scrutiny.

The result is a reactive, fragmented, and increasingly untenable approach to Trust & Safety at enterprise scale.

Why Existing Approaches Fall Short

The Trust & Safety technology market has produced a generation of point solutions—each capable of addressing one dimension of the problem but none capable of delivering the end-to-end, governed investigation workflow that enterprise platforms require.

THE LIMITATIONS OF POINT SOLUTIONS



Content Moderation Tools

Limited to: Text/image flagging only

Missing: Investigation, policy enforcement, auditability



Fraud Detection Systems

Limited to: Transaction pattern analysis

Missing: Behavioral context, policy application, human oversight



Policy Management Platforms

Limited to: Rule definition and tracking

Missing: Evidence gathering, intelligent reasoning, decision automation



Analytics Dashboards

Limited to: Reporting and metrics

Missing: Investigation workflow, decision support, governance



Fragmented workflows, inconsistent decisions, limited visibility, and compliance risk.

TrustOS solves this by unifying investigation, policy, evidence, and governance into a single operating system.

Introducing TrustOS

TrustOS is an **Enterprise AI-Powered Trust & Safety Operating System** that unifies the entire investigation lifecycle—from the moment a marketplace event is flagged to the delivery of a compliance-grade decision and its capture in an auditable execution record.

Unified Investigation Platform
One system orchestrating Trust & Safety from detection to decision.

Multi-Agent AI Intelligence
Specialized AI agents working in concert—not a single black-box model.

Human-in-the-Loop Governance
AI augments human expertise; humans remain decision-makers with structured oversight.

Enterprise-Grade Auditability
Complete decision traceability for compliance and operational transparency.

Rather than bolting together point solutions, TrustOS is purpose-built as an integrated operating system for modern Trust & Safety operations.

Why TrustOS is Different

TrustOS transforms traditional Trust & Safety challenges into strategic advantages, moving beyond reactive, fragmented approaches to deliver an integrated, intelligent, and auditable operating model.

TRADITIONAL TRUST & SAFETY		TRUSTOS
Single AI Model Limited scope, often unable to handle diverse threats.	→	Multi-Agent AI Collaboration Specialized AI agents work in concert for comprehensive investigation.
Black Box Decisions Lack of transparency, difficult to audit or explain.	→	Explainable AI Reasoning Clear, documented reasoning for every decision, enhancing auditability.
Manual Investigation Time-consuming, prone to human error, and inconsistent.	→	AI-Assisted Investigation Autonomous investigations completed in minutes with human-in-the-loop governance.
Static Policy Rules Rigid rules easily circumvented by evolving bad actor tactics.	→	Dynamic Policy Intelligence Adaptive policy enforcement that learns and evolves with new threats.
Fragmented Tools Disparate point solutions create workflow gaps and operational overhead.	↘	Unified Investigation Platform End-to-end workflow orchestrating all aspects of Trust & Safety.
Limited Visibility Lack of real-time insights into investigation patterns and risk trends.	→	Executive Analytics Real-time, actionable insights into platform integrity and operational performance.
Limited Auditability Decisions lack traceability, posing compliance and regulatory risks.	↑	Complete Decision Traceability Full audit log of every investigation, decision, and escalation for regulatory scrutiny.

TrustOS is not another AI assistant.
It is an enterprise operating system purpose-built for Trust & Safety.

Solution Architecture



Marketplace Event

Policy violation detected



TrustOS Investigation Portal

Event enters investigation queue



n8n Orchestration Engine

Workflow coordination



Evidence Agent

Gathers seller history

Transaction patterns

Behavioral signals



Policy Agent

Applies policy rules

Regulatory requirements

Policy citations



Vision Agent

Analyzes images

Authenticity signals

Visual violations



Autonomous Risk Council

Synthesizes agent outputs into unified risk verdict



Compliance Decision Engine

Generates explainable recommendation with confidence score



Human Review

Trust & Safety team makes final decision



Executive Reporter

Creates compliance-grade documentation



Audit Log

Complete traceability

Regulatory evidence



Dashboard & Analytics

Executive visibility

Operational insights

The investigation pipeline transforms marketplace events into governed compliance decisions through specialized AI agents, autonomous risk consolidation, and human-in-the-loop governance. Each component has a defined responsibility and clear handoff, ensuring every investigation is transparent, auditable, and governed by design.

Multi-Agent AI Framework

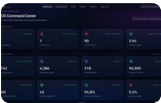
TrustOS distributes investigative responsibility across five specialized AI agents, each designed to perform a specific function with expert-level depth. Rather than relying on a single general-purpose model, this multi-agent architecture enables modularity, transparency, and continuous improvement.

1	Evidence Agent Purpose: Gathers and structures all available evidence Inputs: Seller history, listing content, transaction patterns, behavioral signals, cross-platform signals Outputs: Structured evidence package Consumes: Sellers Intelligence module Key Capability: Foundation for all downstream analysis
2	Policy Agent Purpose: Applies platform policy framework to evidence Inputs: Evidence package, policy repository Outputs: Structured policy assessment with citations Consumes: Policies module Key Capability: Ensures decisions grounded in documented policy
3	Vision Agent Purpose: Performs multi-modal image analysis Inputs: Product images, seller photos, brand assets Outputs: Visual authenticity assessment Consumes: Image data from listings Key Capability: Detects visual policy violations and counterfeits
4	Autonomous Risk Council Purpose: Synthesizes agent outputs into unified verdict Inputs: Evidence assessment, policy assessment, vision assessment Outputs: Unified risk verdict with reasoning chain Consumes: All three specialist agents Key Capability: Resolves conflicts and applies risk weighting
5	Executive Reporter Purpose: Transforms investigation into compliance documentation Inputs: Risk verdict, evidence, policy citations Outputs: Compliance-ready reports and audit log entries Consumes: Risk Council output Key Capability: Produces regulatory evidence and audit trails

The modular design enables each agent to be independently updated, retrained, or replaced as policy evolves or new threat vectors emerge—without requiring a full system rebuild. This modularity is essential for long-term sustainability in a dynamic threat environment.

End-to-End Investigation Workflow & Product Walkthrough

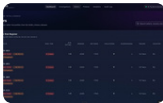
TrustOS delivers a complete, governed investigation workflow from eventing ingestion to decision delivery. Each stage is designed to be transparent, auditable, and operationally efficient.



Module 1: Dashboard

Executive command center for real-time platform monitoring.

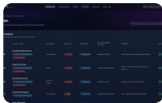
- Live operational metrics (20 active investigations, 1 critical case)
- Real-time KPIs (18,742 listings processed, 4,286 sellers reviewed, 318 violations detected)
- Investigation health indicators (2.8M items scanned, 42 pending reviews, 94.8% accuracy, 5.2% escalation)
- Executive visibility into platform integrity



Module 2: Sellers Intelligence

Comprehensive seller risk profiling and behavioral analysis.

- Seller risk classification (Critical, High, Medium, Low)
- Historical metrics (orders, returns, violations, suspensions)
- Behavioral patterns and marketplace activity tracking
- Evidence Agent uses this data during investigations



Module 3: Policies

Centralized policy repository and enforcement rules.

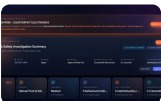
- Policy categories (Counterfeit Electronics, Fake Packaging, Unsafe Electronics)
- Enforcement rules with approval workflows
- Policy status tracking (Active, Inactive, Pending)
- Policy Agent applies these rules during investigation



Module 4: Investigation Queue

Prioritized investigation workflow with SLA tracking.

- Prioritized investigation list with risk scoring
- SLA tracking and status indicators
- AI investigation status (Closed, In Progress, Pending)
- Category-based organization and one-click actions



Module 5: Investigation Detail

AI investigation output with explainable reasoning.

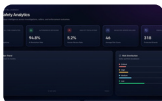
- Automated Trust & Safety Review with AI agent findings
- Investigation summary with agent-level analysis
- Risk assessment and confidence scoring
- Manual review escalation options and recommended next steps



Module 6: AI Investigation Progress

Real-time multi-agent execution monitoring.

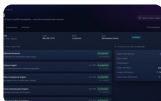
- Real-time agent execution status with progress indicators
- Evidence Agent (gathering marketplace data and seller history)
- Policy Agent (evaluating platform policies)
- Vision Agent, Risk Council, and Executive Reporter execution



Module 7: Analytics

Executive reporting and operational insights.

- Executive KPIs (18,742 investigations, 94.8% accuracy, 5.2% escalation, 46 critical cases)
- Investigation trends and violation patterns over time
- Risk distribution analysis by category
- Top risk indicators and protected items (96,500 items)



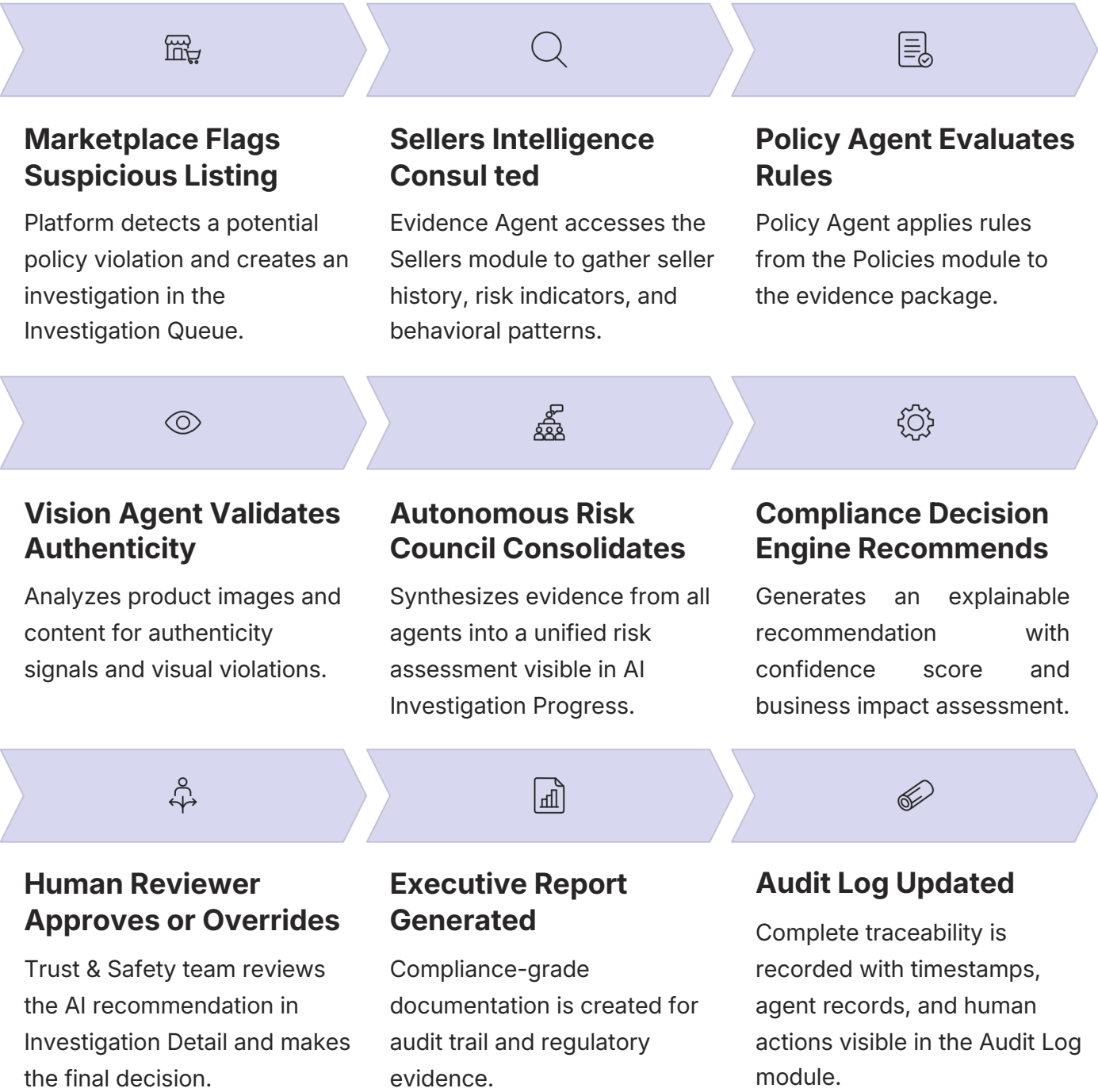
Module 8: Audit Log

Complete compliance evidence and traceability.

- Full investigation traceability with complete history
- Agent execution records (Evidence Agent, Policy Compliance Engine, Brand Authentication Engine, Autonomous Risk Council)
- Timestamp and status tracking for every action
- Compliance status indicators and human oversight records

Example Investigation Journey

This journey illustrates how a real investigation flows through TrustOS, from initial detection through final audit logging. Each step corresponds to actual product modules and demonstrates the collaborative nature of AI agents and human oversight.



Business Value & Responsible AI



Operational Scale

TrustOS enables Trust & Safety teams to operate at scale proportional to platform growth. The MVP demonstrates processing 18,742 investigations with 2.8M items scanned, 4,286 sellers reviewed, and 96,500 items analyzed — all with 94.8% accuracy and only 5.2% escalation rate. AI agents handle high-volume, routine investigations autonomously, freeing human analysts for complex, high-stakes cases.



Decision Consistency

Policy is applied uniformly across every investigation. The platform shows consistent enforcement across categories such as Luxury Goods Detection, Policy Violation, and Counterfeit Detection, with identical violations receiving identical treatment. This eliminates interpretation drift, improves seller trust, and reduces appeals volume.



Regulatory Readiness

TrustOS produces compliance-grade audit trails by design. Every investigation generates documented evidence packages, decision rationale, and audit log entries required by DSA and emerging platform legislation. The Audit Log module provides complete traceability with agent execution records, timestamps, and human oversight documentation.



Executive Visibility

Platform integrity becomes a measurable, reportable operating metric. The Analytics Dashboard provides real-time visibility into Trust & Safety operations with KPIs including investigation trends, risk distribution, violation patterns, and protected items. Executive leadership can report to boards and regulators with confidence.

Responsible AI & Human Oversight

TrustOS is built on a foundational commitment to responsible AI deployment. Automation is deployed where it can be governed, explained, and audited — and human judgment is preserved where it matters most. The platform's human-in-the-loop architecture is not a concession to AI limitations; it is a deliberate design choice reflecting the reality that consequential enforcement decisions affecting livelihoods and platform access carry ethical weight that AI alone should not bear.

Explainability by Design

Every TrustOS decision is accompanied by structured reasoning. The Investigation Output module shows automated Trust & Safety reviews with agent findings, risk assessments, confidence scoring, and recommended next steps. All decisions are grounded in policy citations and evidence references.

Configurable Oversight Thresholds

Platform operators define confidence thresholds for autonomous action versus mandatory human review. The Investigation Queue shows prioritized cases with status indicators (Closed, In Progress, Pending) and one-click escalation actions, giving operators fine-grained control over the automation boundary.

Bias Monitoring & Model Governance

TrustOS includes built-in tooling for monitoring agent decision distributions and detecting bias in enforcement patterns. The Analytics Dashboard shows risk distribution analysis and top risk indicators by category, enabling detection of anomalous decision trends.

Appeals & Contestability

Every autonomous decision can be contested through a structured appeals workflow. The Investigation Output module includes manual review escalation options and recommended next steps, ensuring affected parties have meaningful paths to challenge decisions.

TrustOS Design Principles

TrustOS is engineered from the ground up with a set of core design principles that guide its architecture, features, and operational philosophy. These principles ensure the platform delivers on its promise of ethical, efficient, and auditable Trust & Safety enforcement.



AI Augments, Not Replaces

AI augments—not replaces—human expertise. Trust & Safety professionals remain the decision-makers; AI provides structured reasoning and evidence.



Explainability First

Every recommendation is explainable. No black-box decisions. Every action is traceable to policy, evidence, and reasoning.



Consistent Enforcement

Policy enforcement should be consistent. Identical violations receive identical treatment, reducing appeal friction and bias.



Intentional Oversight

Human oversight is intentional. Not a rubber-stamp process, but a structured review of AI recommendations with clear escalation paths.

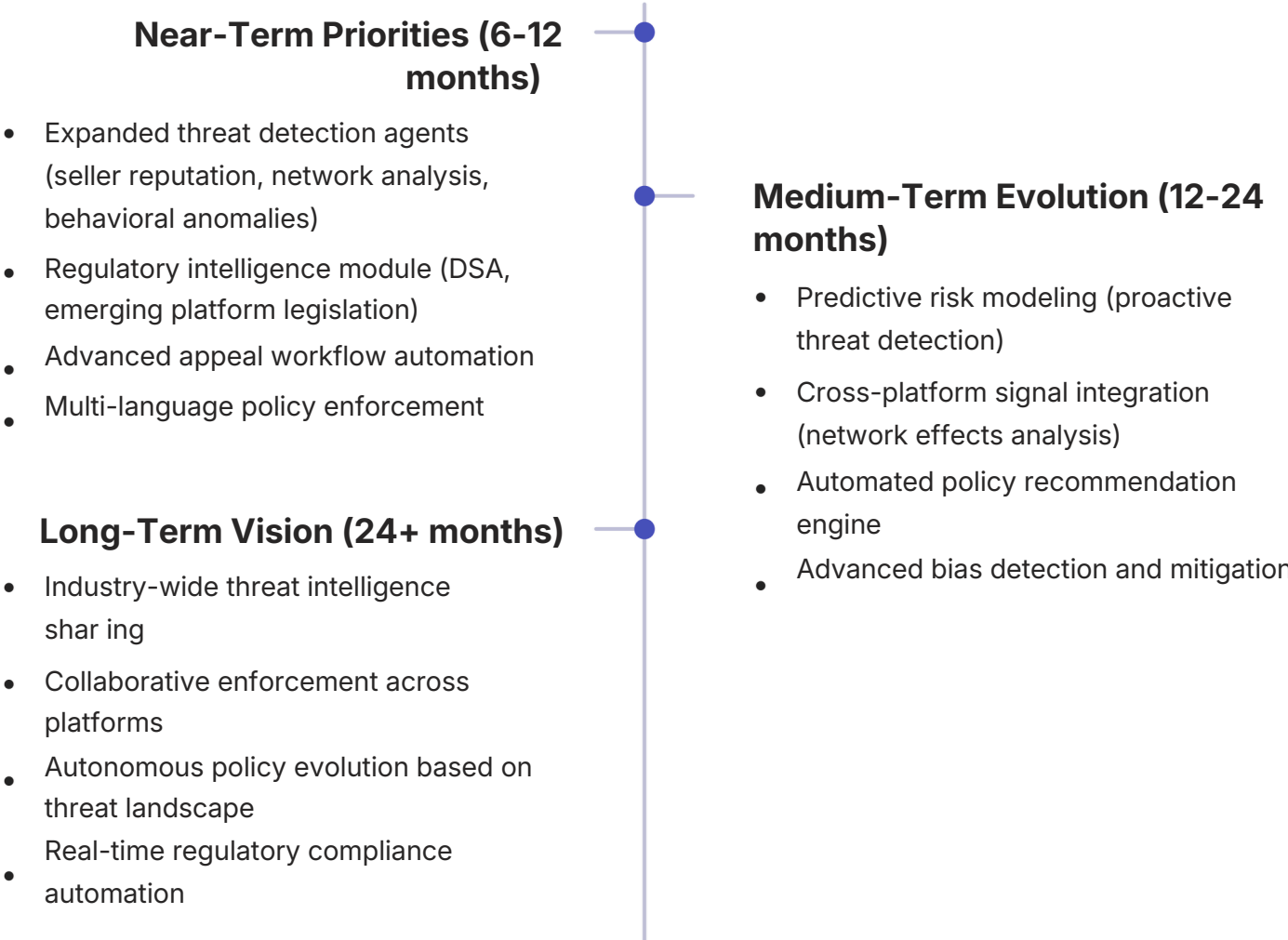


Governance Built-In

Enterprise governance is built into every workflow. Audit trails, compliance documentation, and regulatory evidence are native to the system.

Future Roadmap & Evolution

TrustOS is designed as an evolving platform. The architecture anticipates the expansion of AI capabilities, the evolution of the regulatory landscape, and the emergence of new threat vectors that today's Trust & Safety teams cannot yet predict.



The modular, agent-based architecture ensures TrustOS can evolve without requiring fundamental rebuilds. New agents can be added, existing agents can be enhanced, and the system grows in capability as the threat landscape and regulatory environment evolve.

The Vision

TrustOS demonstrates how collaborative AI can transform Trust & Safety from a reactive moderation function into a proactive, explainable, and intelligence-driven operating capability.

By combining specialised AI agents, transparent decision making, human oversight, and enterprise-grade governance, TrustOS demonstrates how digital platforms can investigate abuse faster while preserving accountability, consistency, and trust.

Rather than replacing human judgement, TrustOS augments it with structured reasoning, policy intelligence, and complete auditability.

This project represents a practical blueprint for the next generation of AI-assisted Trust & Safety operations.

Building safer digital marketplaces through collaborative AI.